SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

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ANSWER SCRIPT EVALUATION SHEET

ASSIGNMENT - I

Programme	B.E / B.Tech	**Course Code**	MLA02
Course Name	**Fundamentals of Machine learning.**	**Date**	21/11/2026
Duration	1 Hr 30 Min	**Max Marks**	50

Register No.: 196526011 6

TASK 1	Max Marks	Marks Awarded
Task 1	5	4
Task 2	2	2
Task 3	6	6
Task 4	4	3
Task 5	3	3
Task 6	3	2½
Task 7	4	3
Task 8	4	3½
Task 9	3	3
Task 10	4	3½
Task 11	4	4
Task 12	8	7
TOTAL	**50**	44½

Total in words:

TASK 1 : Linear Regression using Normal Equations

Given:

x	y
1	2.1
2	3.9
3	6.2
4	7.8
5	10.1

a) Fit $y = w_0 + w_1 x$

The normal equations are

$$w_1 = \frac{n\Sigma xy - \Sigma x \Sigma y}{n\Sigma x^2 - (\Sigma x)^2}$$

and

$$w_0 = \frac{\Sigma y - w_1 \Sigma x}{n}$$

Calculate

$n = 5$, $\Sigma x = 15$, $\Sigma y = 30.1$

$\Sigma x^2 = 55$, $\Sigma xy = 110.7$

Therefore

$$w_1 = \frac{5(110.7) - 15(30.1)}{5(55) - 15^2} = \frac{102}{50} = 2.04$$

$$w_0 = \frac{30.1 - 2.04(15)}{5} = -0.10$$

Hence, the regression equation is

$$\hat{y} = -0.10 + 2.04x$$

b) predict y at $x = 6$

$$\hat{y} = -0.10 + 2.04(6)$$

$$\hat{y} = 12.14$$

c) calculate RSS, MSE & R^2

The predicted values are 1.94, 3.98, 6.02, 8.06, 10.10

Residuals $(y - \hat{y})$ are 0.16, −0.08, 0.18, −0.26, 0

Residual sum of squares: $RSS = \Sigma (y - \hat{y})^2$

$RSS = 0.0256 + 0.0064 + 0.0324 + 0.0676 = 0.132$

Mean squared errors :

$$MSE = \frac{RSS}{n} = \frac{0.132}{5} = 0.0264$$

Total sum of squares:

$$TSS = \sum (y - \bar{y})^2 = 40.268$$

$$R^2 = 1 - \frac{RSS}{TSS} = 1 - \frac{0.132}{40.268} \approx 0.9967$$

(d) Reliability of prediction

The R^2 value indicates an excellent fit to the given data However, $x=6$ is outside the observed range of 1 to 5. Therefore, the prediction is an extrapolation & maybe less reliable If relationship changes beyond available data

Conclusion:

The model fits the given data well, but the prediction at $x=6$ should be treated cautiously

2. TASK 2 : Ridge Regression

Given:

$$S_{xx} = 10 \ , \ S_{xy} = 19.9 \ , \ \lambda = 5$$

The least-squares slope is

$$w_{LS} = \frac{S_{xy}}{S_{xx}} = \frac{19.9}{10} = 1.99$$

The ridge-regression slope is.

$$w_{ridge} = \frac{S_{xy}}{S_{xx} + \lambda}$$

$$w_{ridge} = \frac{19.9}{10+5} = 1.3267$$

Percentage shrinkage

$$\text{Shrinkage} = \frac{w_{LS} - w_{ridge}}{w_{LS}} \times 100$$

$$= \frac{1.99 - 1.3267}{1.99} \times 100$$

Shrinkage $\approx 33.33\%$

Effect of Increasing λ : Increasing λ shrinks coeff toward 0. This increases bias but reduces variance & helps prevent overfitting

TASK 3 : Naiive Bayes Classification

There are 10 emails : 5 spam & 5 Ham

(a) priors & conditional probabilities

Class priors :

$$P(\text{Spam}) = \frac{5}{10} = 0.5$$

$$P(\text{Ham}) = \frac{5}{10} = 0.5$$

conditional probabilities without smoothing :

probability	Spam	Ham
P(offer = Y | class)	4/5 = 0.8	1/5 = 0.2
P(offer = N | class)	1/5 = 0.2	4/5 = 0.8
P(Link = Y | class)	4/5 = 0.8	1/5 = 0.2
P(Link = N | class)	1/5 = 0.2	4/5 = 0.8

(b) classify offer = Y, Link = Y without smoothing using the conditional - independence assumption

$$P(\text{spam} \mid x) \propto P(\text{spam}) P(Y \mid \text{spam}) P(Y \mid \text{spam})$$

Spam score: $0.5(0.8)(0.8) = 0.32$

Ham score: $0.5(0.2)(0.2) = 0.02$

Normalizing : $P(\text{spam} \mid x) = \frac{0.32}{0.32 + 0.02}$

$$P(\text{spam} \mid x) = 0.9412$$

Therefore, email is classified as spam (94.12%)

(c) classification with laplace smoothing, $\alpha = 1$

For a binary feature

$$P(X = v \mid c) = \frac{\text{Count}(X = v, c) + 1}{\text{Count}(c) + 2}$$

for spam : $P(\text{offer}=Y|\text{spam}) = \frac{4+1}{5+2} = \frac{5}{7}$

$P(\text{Link}=Y|\text{spam}) = \frac{5}{7}$

for Ham : $P(\text{offer}=Y|\text{Ham}) = \frac{1+1}{5+2} = \frac{2}{7}$

Spam score: $0.5\left(\frac{5}{7}\right)^2 = \frac{25}{98}$

Ham score: $0.5\left(\frac{2}{7}\right)^2 = \frac{4}{98}$

Therefore: $P(\text{spam}|x) = \frac{25}{25+4}$

$$\boxed{P(\text{spam}|x) = 0.8621}$$

The email is still classified as spam, with posterior probability of approx 86.21%

(d) classify offer=N, Link=N with smoothing

For spam $P(N|\text{spam}) = \frac{1+1}{5+2} = \frac{2}{7}$

for Ham $P(N|\text{Ham}) = \frac{4+1}{5+2} = \frac{5}{7}$

Spam score $0.5 \times \frac{2}{7} \times \frac{2}{7} = \frac{2}{49}$

Ham score $0.5 \times \frac{5}{7} \times \frac{5}{7} = \frac{25}{98}$

Normalizing the scores

$P(\text{spam}|x) = \frac{4}{4+25} = 0.1379$

∴ Email is classified as Ham, with spam probability 13.79%

Analysis: Naive Bayes assumes that offer & link are conditionally independent given the class. This simplifies calculations but may not reflect real email data. Laplace smoothing prevents zero probabilities & makes estimates more robust, although it can reduce extreme probability estimates.

Task 4: Logistic Regression - One gradient descent step

Given: $w_1 = w_2 = b = 0$, $\eta = 0.5$

The sigmoid function is: $\boxed{p = \frac{1}{1+e^{z}}}$, $z = w_1 x_1 + w_2 x_2 + b$

Initially, both samples have $p = 0.5$.

Step 1 - Calculate the gradient for mean cross-entropy loss:

$$\frac{\partial L}{\partial w_j} = \frac{1}{2}\sum_{i=1}^{2}(p_i - y_i)x_{ij}$$

$$\frac{\partial L}{\partial b} = \frac{1}{2}\sum_{i=1}^{2}(p_i - y_i)$$

for w_1: $\frac{\partial L}{\partial w_1} = \frac{(-0.5)(1) + (0.5)(2)}{2} = 0.25$

for w_2: $\frac{\partial L}{\partial w_2} = \frac{(-0.5)(2) + (0.5)(1)}{2} = -0.25$

for the bias: $\frac{\partial L}{\partial b} = \frac{-0.5 + 0.5}{2} = 0$

Hence gradient is

$$\boxed{\nabla L = (0.25, -0.25, 0)}$$

Step 2: update the parameters using $\theta_{new} = \theta_{old} - \eta \nabla L$:

$w_1 = 0 - 0.5(0.25) = -0.125$

$w_2 = 0 - 0.5(-0.25) = 0.125$

$b = 0 - 0.5(0) = 0$

$\therefore \boxed{w = (-0.125, 0.125),\ b = 0}$

Step 3: predicted probabilities & loss

For sample 1, $x = (1, 2)$

$z = -0.125(1) + 0.125(2) = 0.125$

$p_1 = \sigma(0.125) \approx \boxed{0.5312}$

for sample 2, $x=(2,1)$: $z=-0.125(2)+0.125(1)=-0.125$

$p_2=\sigma(-0.125)\approx \boxed{0.4688}$

Initial mean loss: $h_{before}=-\ln(0.5)=\boxed{0.6931}$

updated mean loss: $h_{after}=\frac{-\ln(0.5312)-\ln(1-0.4688)}{2}\approx \boxed{0.6320}$

Analysis:

The loss decreases from 0.6931 to 0.6320. Thus, the gradient descent step improves the model by increasing the probability of the +ve sample & decreasing that of -ve sample

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5. TASK 5: Entropy, Information gain & Gini impurity

Given: the split produces:

$A=d_1$: 4 yes, 0 No.

$A=d_2$: 2 yes, 4 No.

Step 1 - Entropy of parent node

$H(S)=-P_+\log_2 P_+ - P_-\log_2 P_-$

$H(S)=-0.6\log_2(0.6)-0.4\log_2(0.4)$

$\boxed{H(S)=0.971 \text{ bits}}$

Step 2 - Information gain

for d_1, all samples are yes: $H(d_1)=0$

for d_2 $H(d_2)=-\frac{2}{6}\log_2\frac{2}{6}-\frac{4}{6}\log_2\frac{4}{6}$

$H(d_2)=0.9183$

weighted entropy after splitting

$H(S|A)=\frac{4}{10}(0)+\frac{6}{10}(0.9183)=0.5510$

$\therefore \boxed{IG=H(S)-H(S|A)}=0.971-0.551=\boxed{0.420}$

Step 3 - Gini impurity decrease

parent Gini impurity:

$G(S)=1-(0.6^2+0.4^2)=0.48$

for d_1: $G(d_1)=0$

for d_2: $G(d_2)=1-\left[\left(\frac{2}{6}\right)^2+\left(\frac{4}{6}\right)^2\right]=0.4444$

ighted child impurity:

G children $= \frac{4}{10}(0) + \frac{6}{10}(0.4444) = 0.2667$

Gini decrease $= 0.48 - 0.2667 = 0.2133$

Conclusion: The split is worthwhile because both IG & Gini impurity decrease are positive, indicating improved class seperation.

TASK 6 : Cost - Complexity pruning

Given:

Training errors of subtree : $R(T) = 3$

No. of subtree leaves: $[T] = 4$

Errors after collapsing to one leaf : $R(t) = 8$

The cost-complexity fcn is : $R_\alpha(T) = R(T) + \alpha|T|$

Subtree cost: $R_\alpha(T) = 3 + 4\alpha$

Single leaf cost: $R_\alpha(t) = 8 + \alpha$

Step 1 - Find the critical value

At equal costs : $3 + 4\alpha = 8 + \alpha$

$3\alpha = 5$

$\alpha = \frac{5}{3} \approx 1.667$

Step 2 - Evaluate the pruning decision

For $\alpha = 0.01$: $R_\alpha(T) = 3.04$, $R_\alpha(t) = 8.01$

The subtree has lower cost, so do not prune.

for $\alpha = 0.02$: $R_\alpha(T) = 3.08$, $R_\alpha(t) = 8.02$

Again, the subtree has lower cost, so do not prune.

Conclusion:

The subtree should be retained for both given values of α. pruning becomes preferable when α exceeds the critical value of approximately 1.667

7. TASK 7 - Neutral Networks & Back propagation

Given: $h = \sigma(w_1 x_1 + w_2 x_2)$, $\hat{y} = \sigma(vh)$

$w_1 = 0.4$, $w_2 = 0.3$, $v = 0.5$

$x = (1, 2)$, $t = 1$, $\eta = 0.5$

The sigmoid function is: $\sigma(z) = \frac{1}{1+e^{-z}}$

Step 1 : Forward pass

Hidden unit:

$z_h = 0.4(1) + 0.3(2) = 1$

$h = \sigma(1) = \boxed{0.7311}$

Output unit:

$z_o = 0.5(0.7311) = 0.3655$

$\hat{y} = \sigma(0.3655) = \boxed{0.5904}$

Squared-error loss

$E = \frac{1}{2}(t - \hat{y})^2$

$E = \frac{1}{2}(1 - 0.5904)^2 = \boxed{0.0839}$

Step 2 : Backpropagation error terms

Output error term, using the target-minus-output convention

$\delta_o = (t - \hat{y})\,\hat{y}(1 - \hat{y})$

$\boxed{\delta_o = 0.09906}$

Hidden error term:

$\delta_h = h(1-h)v\delta_o$

$\boxed{\delta_h = 0.009738}$

Step 3 : Calculate gradients

$\frac{\partial E}{\partial v} = -\delta_o h = \boxed{-0.07242}$, $\frac{\partial E}{\partial w_1} = -\delta_h x_1 = \boxed{-0.009738}$

$\frac{\partial E}{\partial w_2} = -\delta_h x_2 = \boxed{-0.01948}$

Step 4 : update weights

using $w_{new} = w_{old} - \eta \frac{\partial E}{\partial w}$

$v = 0.5 + 0.5(0.07242) = 0.53621$

$w_1 = 0.4 + 0.5(0.009738) = 0.40487$

$w_2 = 0.3 + 0.5(0.01948) = 0.30974$

5: calculate the new loss

hidden activation $h' = \sigma(0.40487) + 2(0.30974) \approx 0.7358$

new output $\hat{y}' = \sigma(0.53621 \times 0.7358) \approx 0.5974$

$E' = \frac{1}{2}(1 - 0.5974)^2$

$E' = 0.0811$

conclusion:

The loss decreases from 0.0839 to 0.0811. Backpropagation updates the weights to reduce prediction error.

TASK 8: Support Vector Machine (SVM)

Given: $w = (0.5, 0.5)$, $b = -2$

Decision function: $f(x) = w^T x + b = 0.5x_1 + 0.5x_2 - 2$

a) verify support vectors & calculate margin width

for (3,3) class +1 $f(3,3) = 1.5 + 1.5 - 2 = 1$

for (1,1) class -1 $f(1,1) = 0.5 + 0.5 - 2 = -1$

Both support vectors lie on the margin hyperplanes $f(x) = +1$ & $f(x) = -1$

The norm of the weight vector is:

$$\|w\| = \sqrt{0.5^2 + 0.5^2} = \sqrt{0.5}$$

Margin width: $\frac{2}{\|w\|} = \frac{2}{\sqrt{0.5}} = 2.828$

(b) classify points & calculate distances

The distance from a point to decision boundary is:

$$d = \frac{|f(x)|}{\|w\|}$$

for (4,2): $f(4,2) = 2 + 1 - 2 = 1$

class (+1): $d = \frac{1}{\sqrt{0.5}} = 1.414$

for (1,2): $f(1,2) = 0.5 + 1 - 2 = -0.5$

class (-1): $d = \frac{0.5}{\sqrt{0.5}} = 0.707$

(c) Hinge loss & margin violations

Hinge loss is: $L = \max(0, 1 - y f(x))$

for (1,2) true label $y = -1$ $L = \max(0, 1-(-1)(-0.5)) = 0.5$

for (2,2) true label $y = +1$ $L = \max(0, 1-1(0)) = 1$

3 Conclusion: Both points violate the unit-margin requirement because their hinge losses are +ve However both are classified correctly acc. to their respective labels.

9. TASK-9 : Ensemble Learning using Majority Voting

Given: Accuracy of each classifier $p = 0.70$, $q = 0.30$

(a) Accuracy for 3 classifiers

$$P(3) = \binom{3}{2} p^2 q + p^3$$

$$= 3(0.7)^2(0.3) + (0.7)^3$$

$$P(3) = 78.4\%$$

(b) Accuracy for 5 classifiers

$$P(5) = \binom{5}{3} p^3 q^2 + \binom{5}{4} p^4 q + p^5$$

3 $$= 10(0.7)^3(0.3)^2 + 5(0.7)^4(0.3) + (0.7)^5$$

$$P(5) = 83.69\%$$

(c) Analysis:

Accuracy increases from 78.4% to 83.69% when two classifiers are added. If errors are strongly correlated, the improvement decreases because classifiers tend to make the same mistakes.

TASK 10 : AdaBoost Algorithm

Given: 5 equally weighted samples. samples 2 & 5 are misclassified

Step 1 - weighted error

Initial weight of each sample : $w_i = \frac{1}{5} = 0.2$

$\epsilon = 0.2 + 0.2 = 0.4$

Step 2: Learner weight

$$\alpha = \frac{1}{2} \ln\left(\frac{1-\epsilon}{\epsilon}\right) = \frac{1}{2} \ln\left(\frac{0.6}{0.4}\right) = 0.2027$$

Step 3: update & normalise weights

$$w_i' = w_i e^{-\alpha y_i h_i}$$

correctly classified samples: $0.2 e^{-0.2027} = 0.1633$

Misclassified samples: $0.2 e^{0.2027} = 0.2449$

After normalisation:

Sample	Final weight
1 (correct)	0.1667
2 (Incorrect)	0.2500
3 (correct)	0.1667
4 (correct)	0.1667
5 (Incorrect)	0.2500

Conclusion:

AdaBoost increases the weights of misclassified samples and decreases the weights of correctly classified samples. Consequently, the next weak learner focuses more on difficult samples

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TASK 11 : Discriminant functions

Given: $\Sigma = \begin{bmatrix} 2 & 0.5 \\ 0.5 & 1 \end{bmatrix}$, $\mu_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\mu_2 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$

a) Discriminant direction & threshold:

$$\Sigma^{-1} = \frac{1}{1.75}\begin{bmatrix} 1 & -0.5 \\ -0.5 & 2 \end{bmatrix}, \quad w = \Sigma^{-1}(\mu_2 - \mu_1)$$

$$\boxed{w = \begin{bmatrix} 1.4286 \\ -1.7143 \end{bmatrix}}$$

for equal priors: $c = \frac{1}{2}(\mu_1 + \mu_2)^T w$

$$\boxed{c = 0.2857}$$

b) Classify $x = (2.5, 4)$

$$w^T x = 1.4286(2.5) - 1.7143(4) = 1.8572$$

Since $1.8572 > 0.2857$, the point belongs to class w_2.

c) unequal priors

Given $P(w_1) = 0.8$, $P(w_2) = 0.2$;

$$c' = c + \ln\frac{P(w_1)}{P(w_2)}$$

$$c' = 0.2857 + \ln 4 = 1.6720$$

Since $1.8572 > 1.6720$

The point still belongs to class w_2.

Conclusion:

unequal priors change the threshold, but the classification remains unchanged.

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12. TASK 12 : Evaluation of Bank Loan Classifiers

Given: 1000 loans, 100 actual defaulters. FN cost = ₹ 50,000 & FP cost = ₹5000.

(a) Calculate evaluation metrics

$$\text{Accuracy} = \frac{TP+TN}{1000} \times 100$$

$$\text{precision} = \frac{TP}{TP+FP} \times 100$$

$$\text{Recall} = \frac{TP}{TP+FN} \times 100$$

$$F1 = \frac{2TP}{2TP+FP+}$$

Metric

Accuracy

precision

Recall

F1-score

(b) Total

Model A

Model B

Model

(c) Sele

Model

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$= \frac{2TP}{2TP+FP+FN} \times 100$

Metric	Model A	Model B	Model C
Accuracy	88%	80.5%	93%
precision	43.75%	32.08%	66.67%
Recall	70%	85%	60%
F1-score	53.85%	46.58%	63.16%

(b) Total expected cost

$$\text{Cost} = (FN \times 50000) + (FP + 5000)$$

Model A - 30(50000) + 90(5000) = ₹ 19,50,000

Model B - 15(50000) + 180(5000) = ₹ 16,50,000

Model C - 40(50000) + 30(5000) = ₹ 21,50,000

(c) Select the best model

Model B is preferred because it has the lowest total cost of ₹16,50,000 & the highest recall (85%), Model C has the highest accuracy and F1 score, but misses more defaulters, increasing financial losses.

(d) Improvements & validation

1. Adjust the classification threshold to minimise total cost.
2. use class-weighted learning to improve default detection.
3. Apply cross-validation & test the final model on seperate unseen data to prevent overfitting

Conclusion:

Model B is the best choice based on the given financial costs.